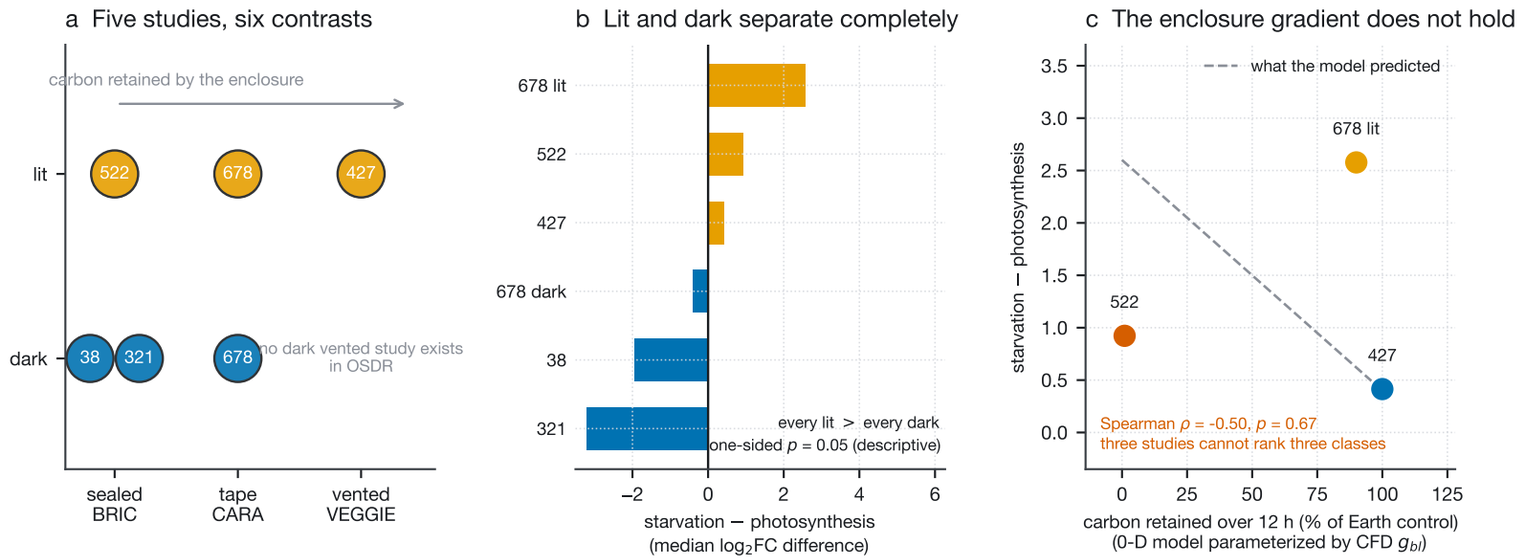
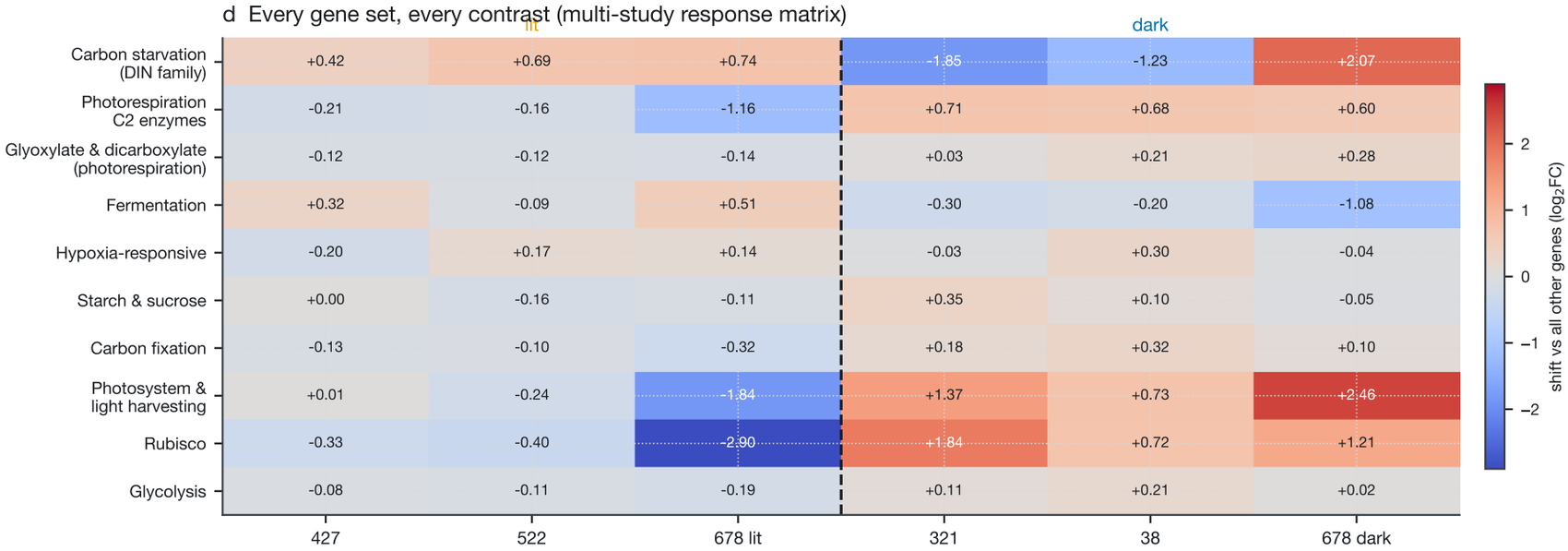
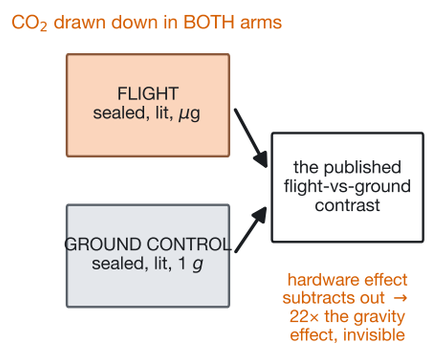
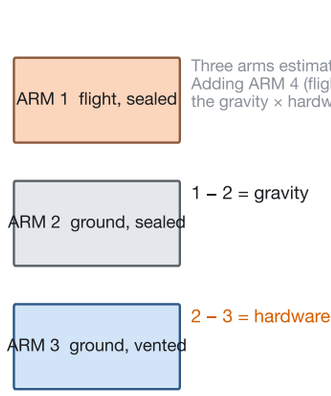
# Cross-study evaluation across six OSDR flight contrasts and experimental design to unconfound hardware from microgravity




## g The NASA OSDR archive gap

**0 of 12** OSDR metabolomics studies that are of a plant

**12 of 567** OSDR studies that are metabolite profiling, all mouse, human or microbe

**0 known** in-canister $CO_2$ logs, any BRIC-class flight

A ~20-compound targeted panel on archived material would be the single most informative experiment.

Six matched flight-vs-ground contrasts from NASA OSDR; gene sets from KEGG; model from LunarLeaf-CFD & FvCB.